FORM-3D:
Fast **O**ptimization of **R**ange **M**odulators **3D**

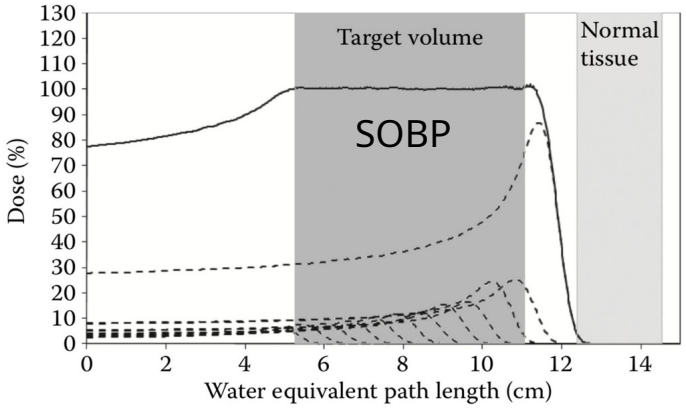
Grant Giovani INFN, November, 19th, 2024

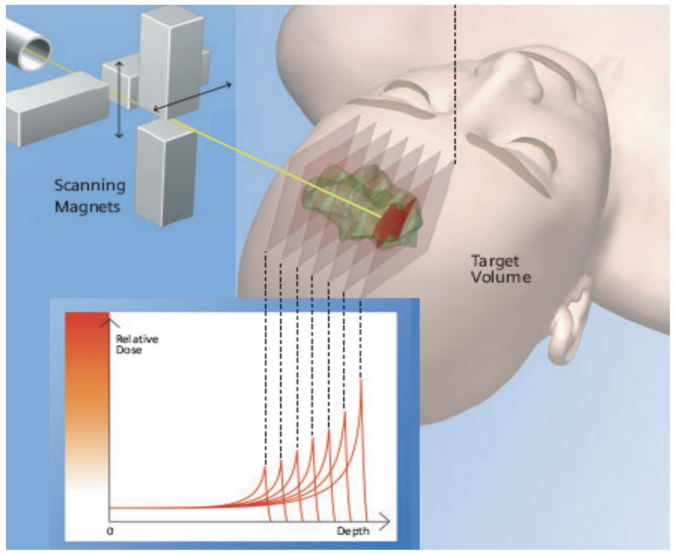
Alberto Taffelli

Rome, November, 19th, 2024

INTRODUCTION

Proton Therapy:
Protons can deliver dose precisely in deep seated tumours thanks to their peaked dose distribution in depth (Bragg peak), thus **sparing normal tissues**





Active Pencil beam scanning (PBS) systems make use of **different energy layers** to reproduce dose uniformity in the target (SOPB)

COST FUNCTION

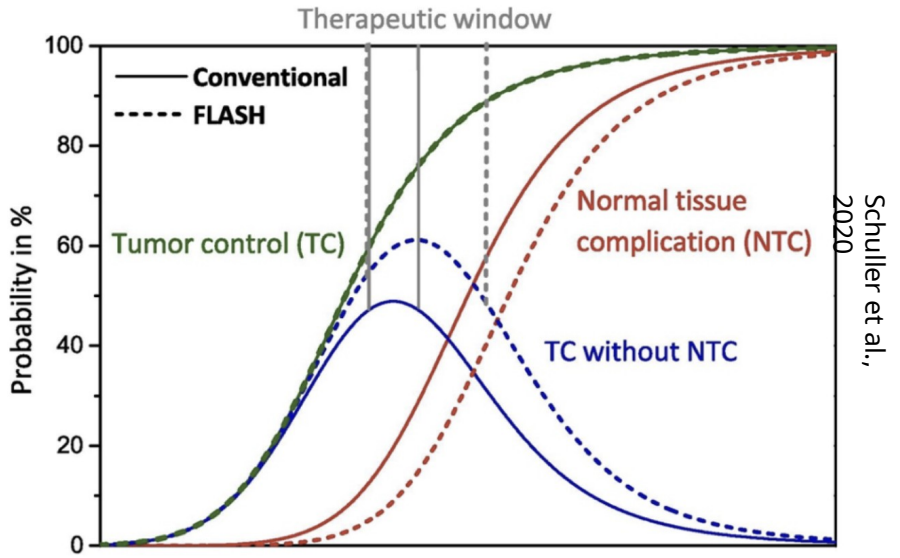
$$\chi^2(\vec{N}) = (w_t)^2 \sum_{i=1}^{N_T} \frac{\left(D_{pre} - D_i(\vec{N})\right)^2}{\Delta D_{pre}^2} + \left(w_{OAR}^{Dmax}\right)^2 \sum_{i=1}^{N_{OAR}^{Dmax}} \frac{\left(D_{max} - D_i(\vec{N})\right)^2}{\Delta D_{max}^2} \cdot \theta\left(D_i(\vec{N}) - D_{max}\right)$$

$$\{E_1, \dots, E_n, \sigma_1, \dots, \sigma_n\}$$

n ∽ 10^3-10^4

FLASH Effect:
Ultra-high dose rates (UHDR) irradiation, i.e. >40 Gy/s, have a protective effect on normal tissues, compared to conventional irradiation (0.01Gy/s), while maintaining the same efficacy against tumours.

In RT = increase the **therapeutic windo…**

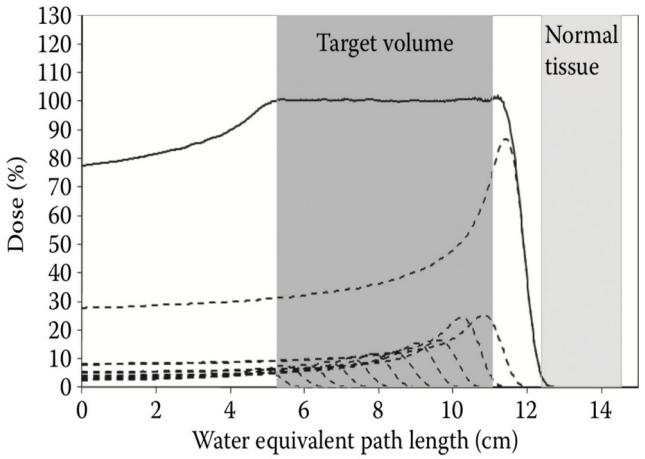


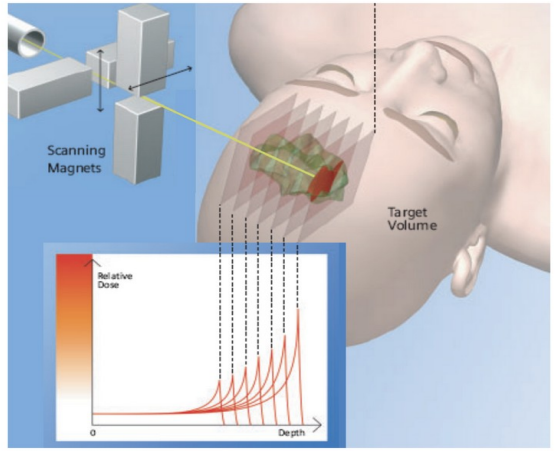
INTRODUCTION

FLASH Effect in proton therapy:

Problem: modern PBS proton therapy centers (PTC) can't deliver UHDR due to the energy switching time required between different energy layers (0.5-2 s)

The FLASH effect is lost

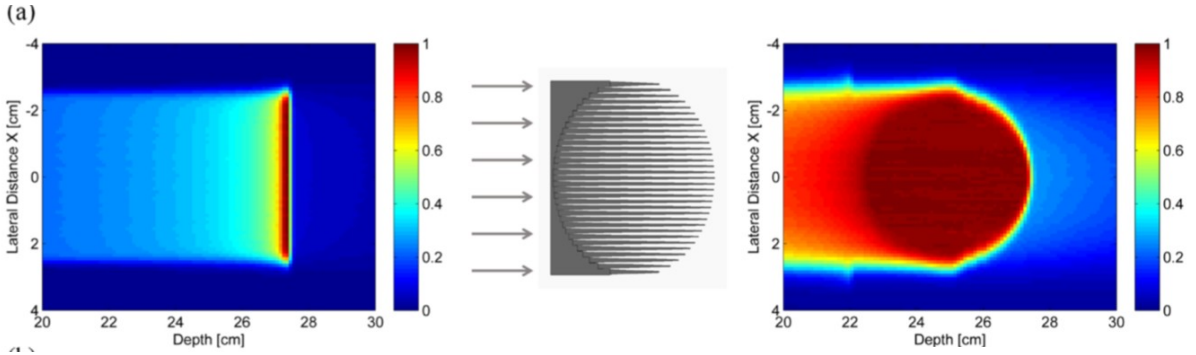




3D Range Modulators (3DRM)

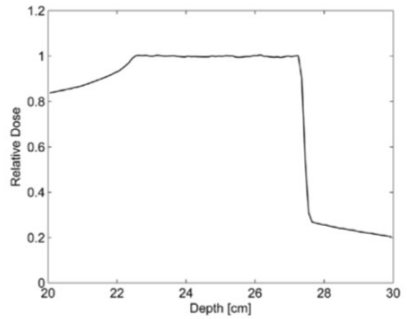
Solution: 3DRM, objects able to passively modulate the proton pencil beam energy and conformally shape the dose in the target (tumor)

- Optimization based on pre-calculated look-up-tables
- Each pin is designed to contribute correctly to the final dose distribution (pin's lengths)
- Fluence of e eds to be optimized $\{l_1^1, \dots, l_i^k, \sigma_1, \dots, \sigma_n\}$



Simoneov et al. 2017





PURPOSE

Fast **O**ptimization of **R**ange **M**odulators **3D**

Why?

- The design of 3DRM is a non-trivial optimization process, dealing with **inaccurate 3D look-up-tables,** requiring **long computational times**, **uncompatible with clinical routine** (up to days)
- A complete workflow to design, print and test the 3DRM is still missing

To speed up the translation of FLASH proton therapy into clinics

How?

- Exploiting the **superior computational speed of FRED**, a GPU-based MC simulation tool, in the optimization process
- Creating a dedicated **efficient optimization algorithm** to design the 3DRM
- Developing a **complete workflow** for the design, printing and

Copyright © 2015-2023. A. Schiavi and Sapienza university of Rome. All rights reserved

	Hardware	primary/s	Patient plan recalculation*
FLUKA/GEANT4	single CPU card	750	16 days
FRED	single CPU card	15000	19 hours
FRED	single GPU card	10 mln	2.3 min
FRED	cluster of 144 GPU cards	300 mln	3 s

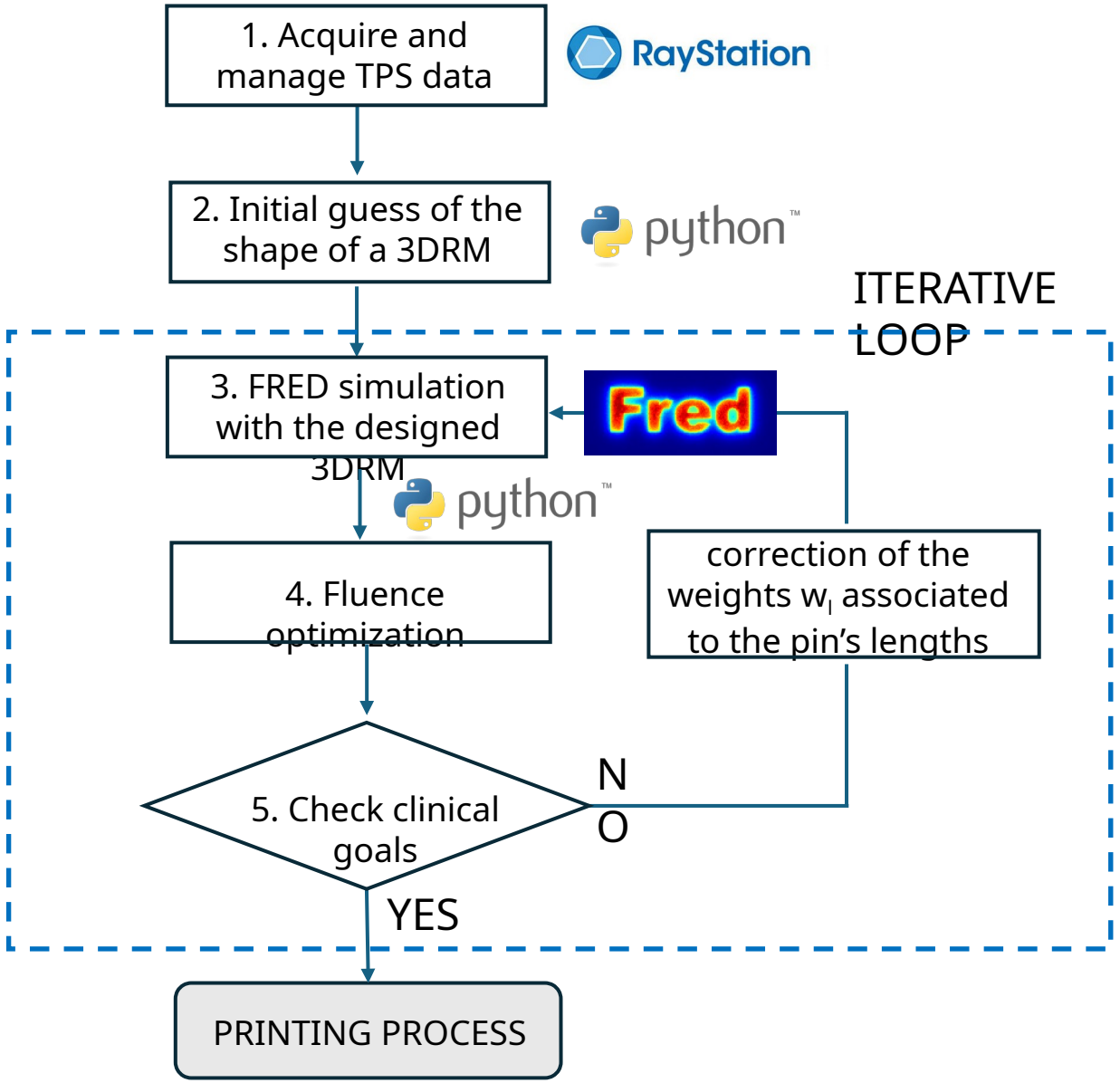
www.fred-mc.org

FRED EXPERTISE:

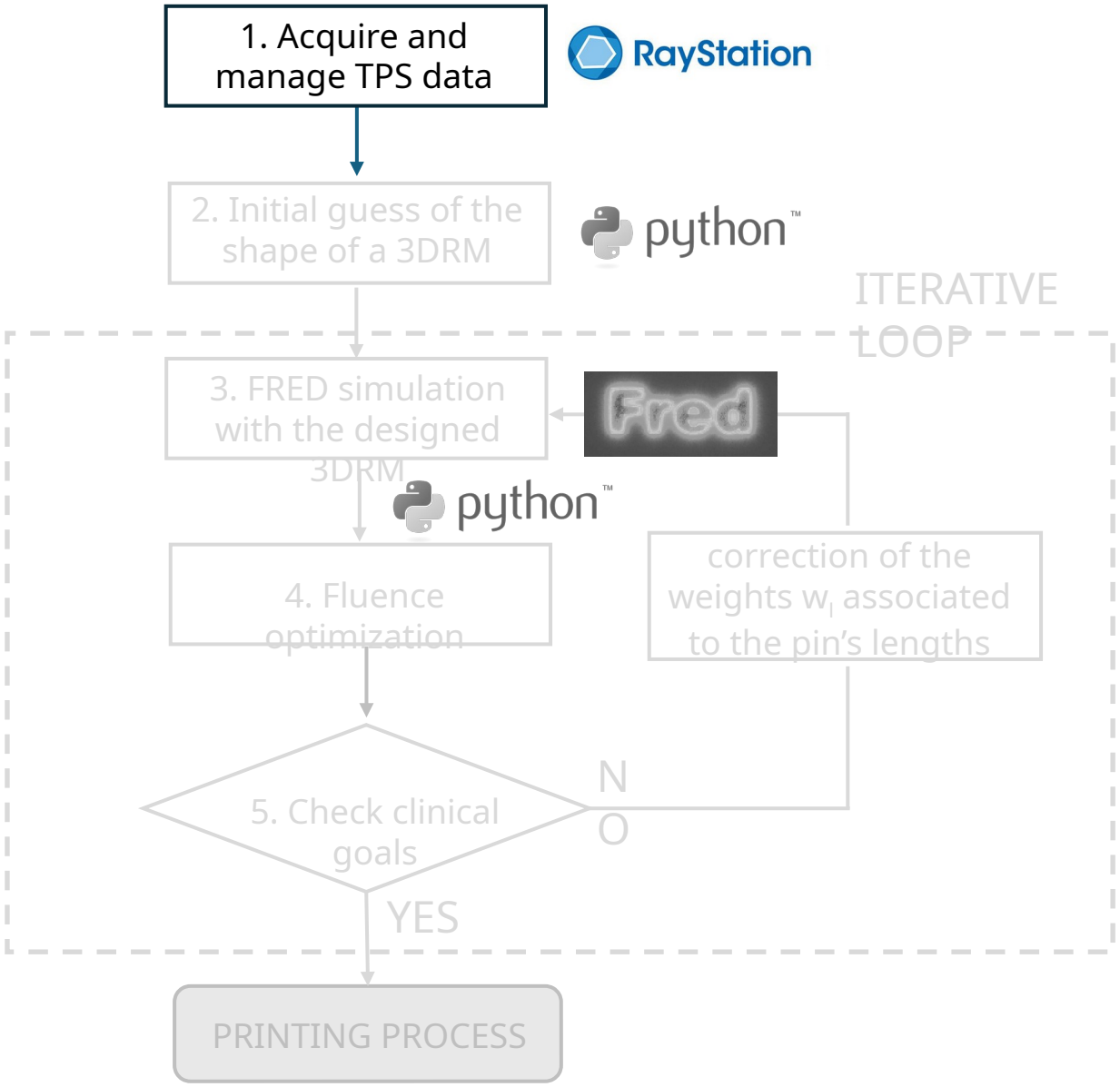
FRED Commissioning during FRIDA:

- APSS beam model implemented in FR ✓
- Tool to recalculate patient's RT plan developed ✓
- Validated on a set of significant patie ✓ ;

3DRM DESIGN

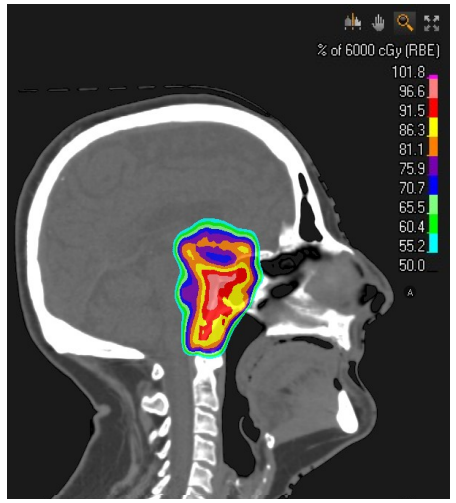

1. Acquire and manage TPS data
RayStation
2. Initial guess of the shape of a 3DRM
python
ITERATIVE LOOP
3. FRED simulation with the designed 3DRM
Fred
python
4. Fluence optimization
correction of the weights w_l associated to the pin's lengths
5. Check clinical goals
NO
YES
PRINTING PROCESS

3DRM DESIGN



TPS dicom files:

- RT plan
- RT dose
- CT
- Structures



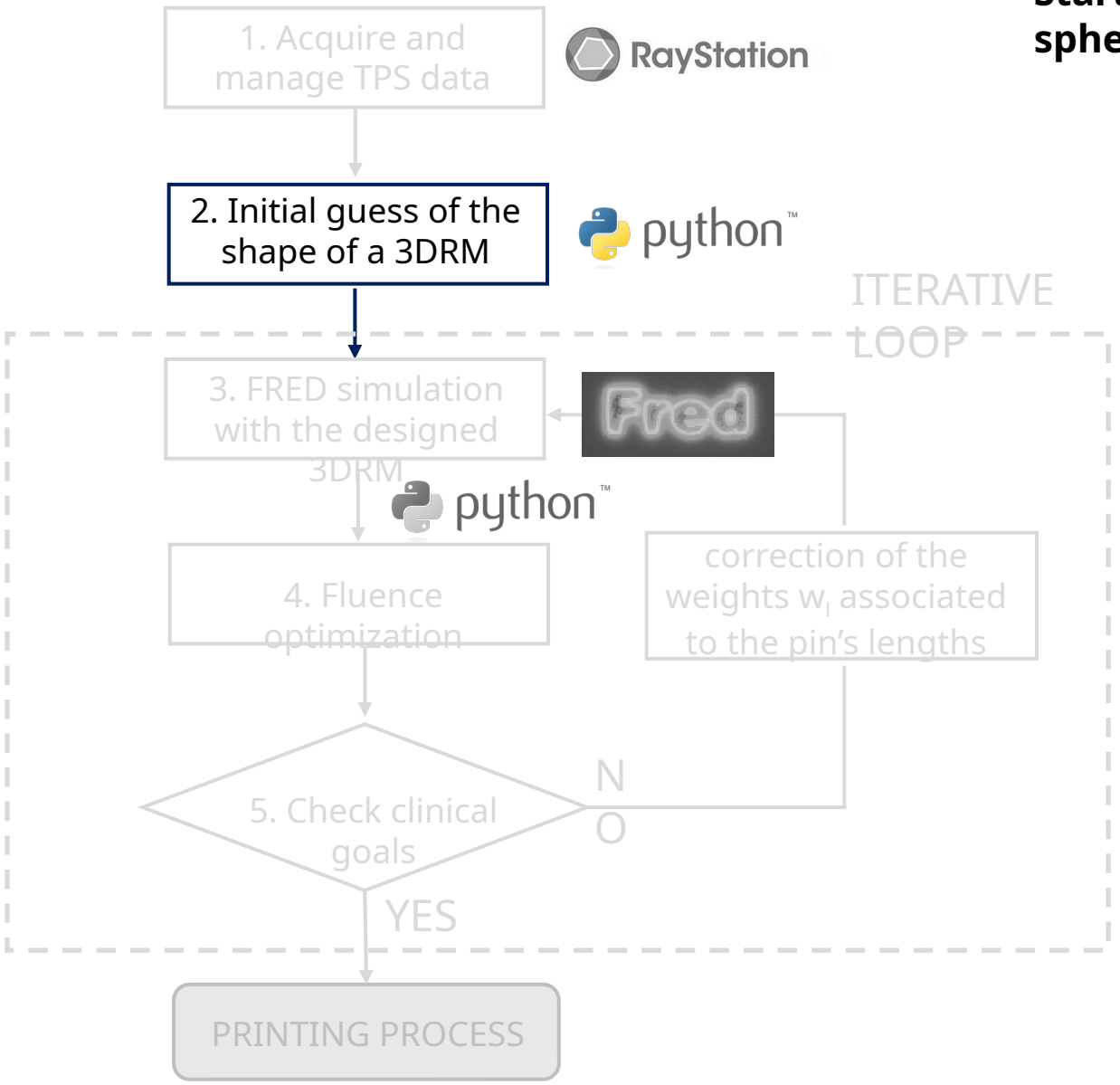
RT original plan

```
pb: 001 1; E=105.6;
pb: 002 1; E=105.6;
…
pb: 301 1; E=98.5;
pb: 302 1; E=98.5;
…
pb: 701 1; E=78.3;
pb: 702 1; E=78.3;
…
```

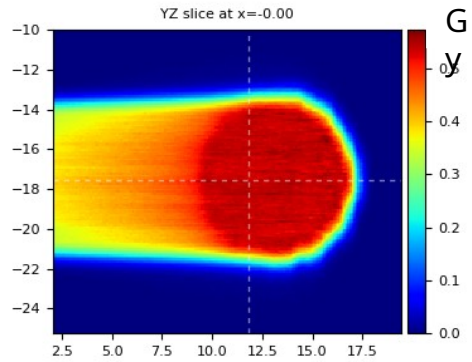
3DRM based plan

```
pb: 001 1; E=105.6;
pb: 002 1; E=105.6;
…
pb: 301 1; E=105.6;
pb: 302 1; E=105.6;
…
pb: 701 1; E=105.6;
pb: 702 1; E=105.6;
…
```

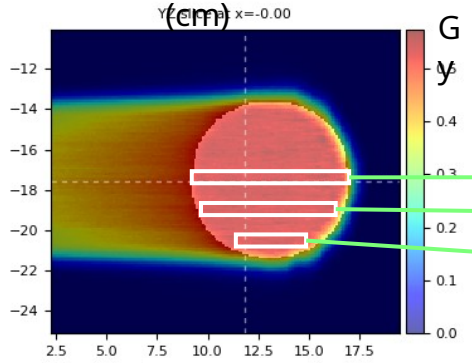
3DRM DESIGN



Starting from simple geometries (e.g. sphere)

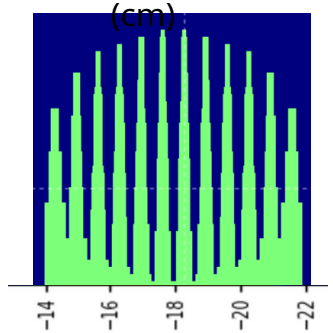


TPS dose distributio n



Target structure (e.g. PTV)



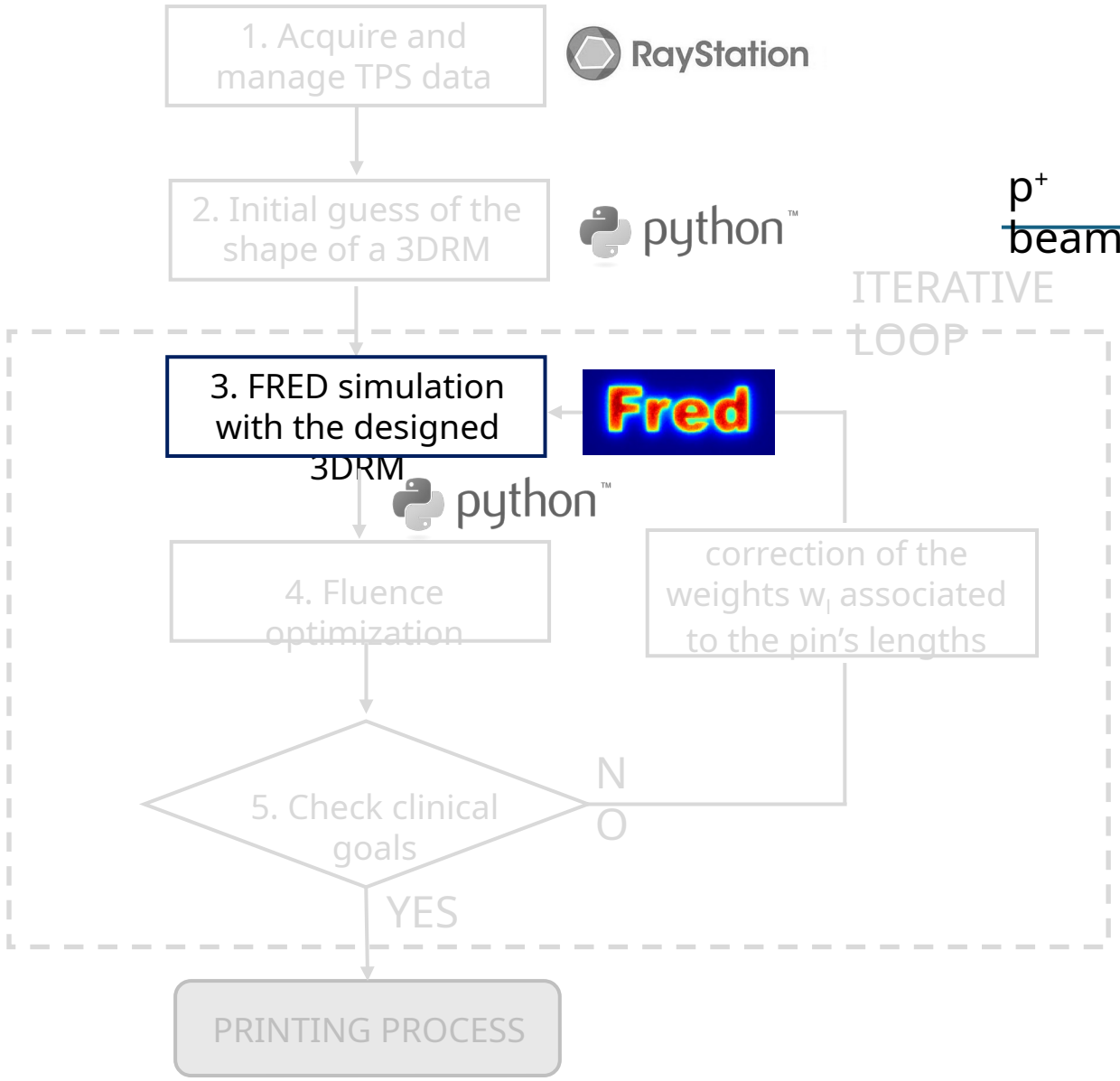


Raw 3DRM

Quick guess of the 3DRM based on:

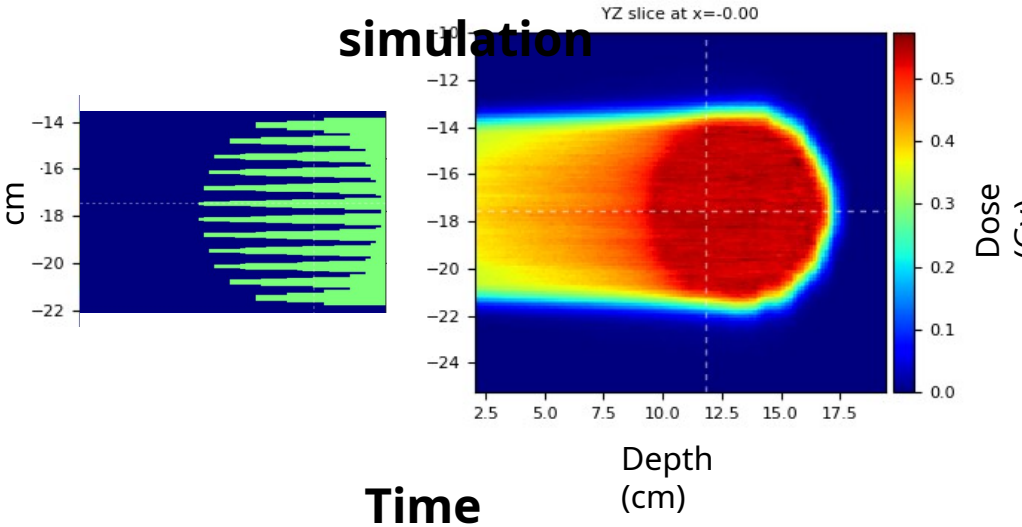
- Geometrical considerations on the target
- Pre-optimized independent 2D SOPBs

3DRM DESIGN



p^+ beam

FRED simulation

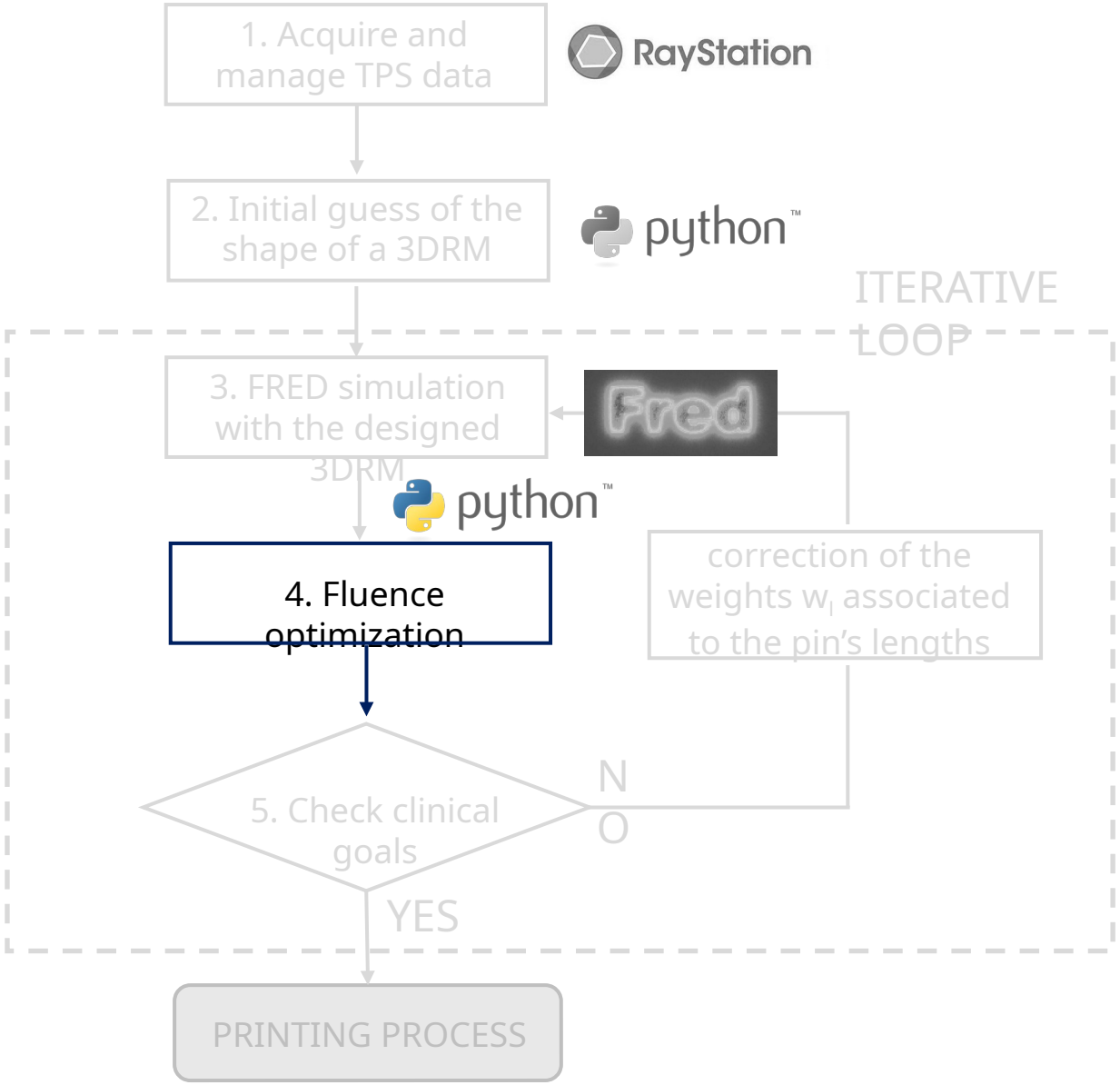


Time estimate

RM #vox	Phase Space #vox	nPrim	FRED Time
10^6	10^6	10^5	2 s
10^6	10^6	10^6	2 s
10^6	10^6	10^7	4 s
6.4 x 10^7	10^6	10^5	3 s
6.4 x 10^7	10^6	10^6	3 s
6.4 x 10^7	10^6	10^7	7 s
10^9	10^6	10^5	20 s
10^9	10^6	10^6	15 s
10^9	10^6	10^7	23 s

Very fast recalculation!

3DRM DESIGN



Fluence Optimization

$$D_i = \sum_{j=1}^{N} w_j * d_{ij}$$

python

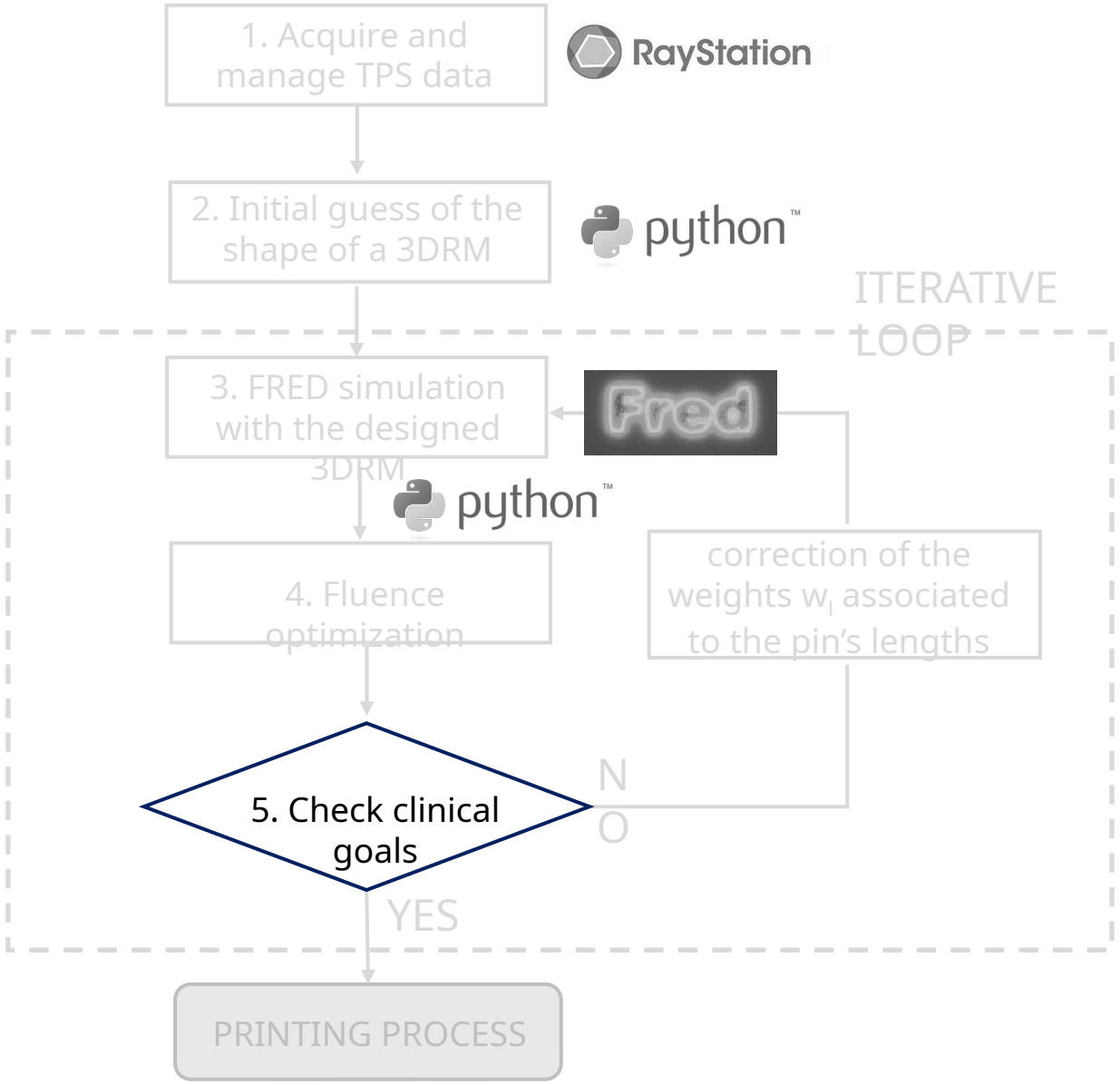
dose in the ith voxel

dose influence matrix

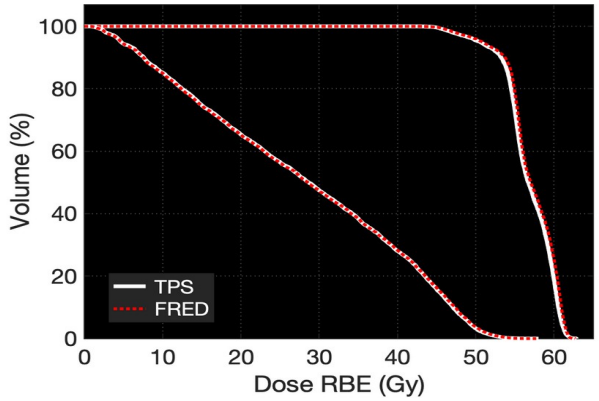
weight of the jth pb (fluence)

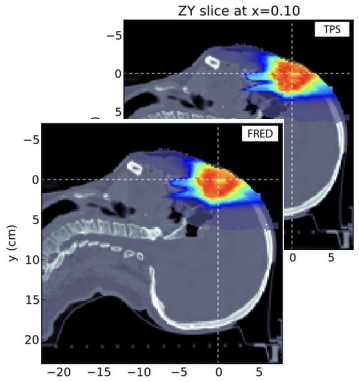
optimize the weights w_j to converge to the objective dose distribution D_i
e.g. (PTV coverage, OARs sparing)

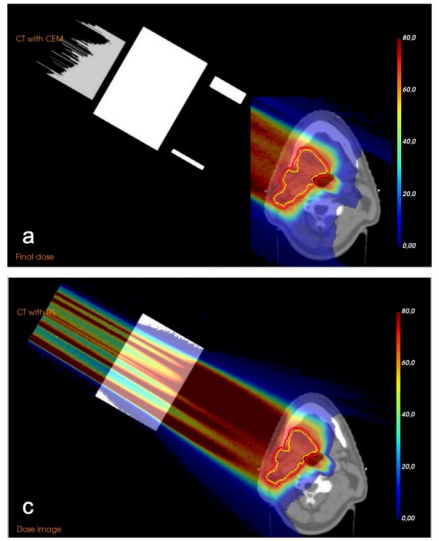
3DRM DESIGN



Compare DVH metrics: D98%, D2%, Dmax

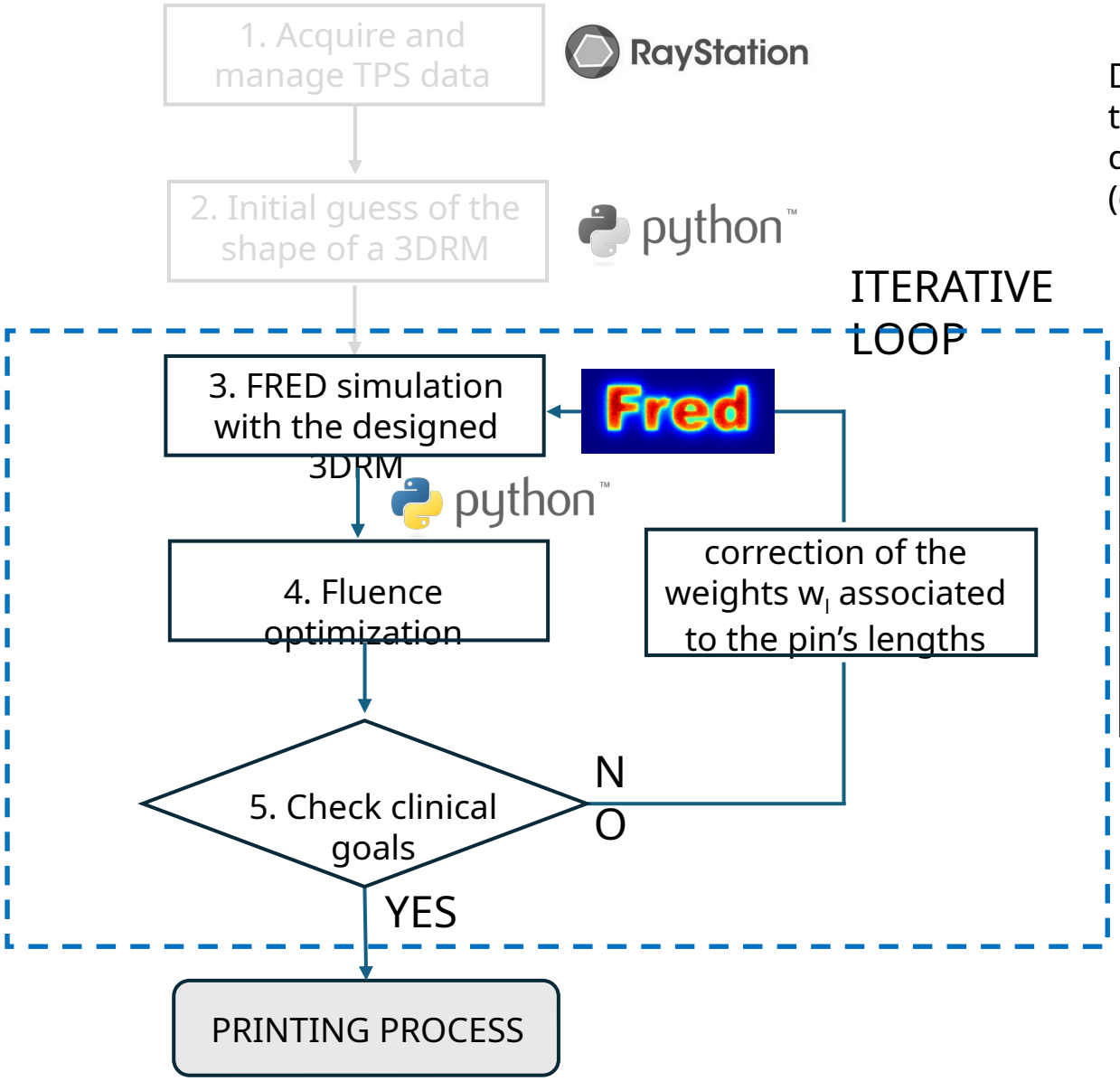




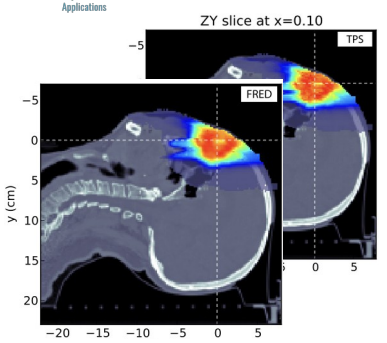


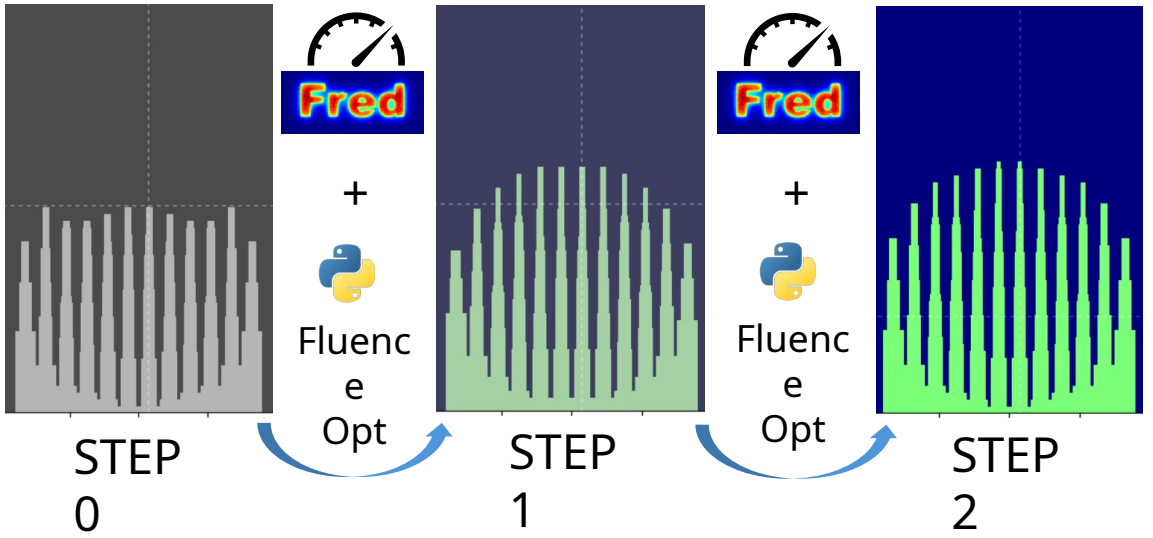
Deffet et al., 2023

3DRM DESIGN



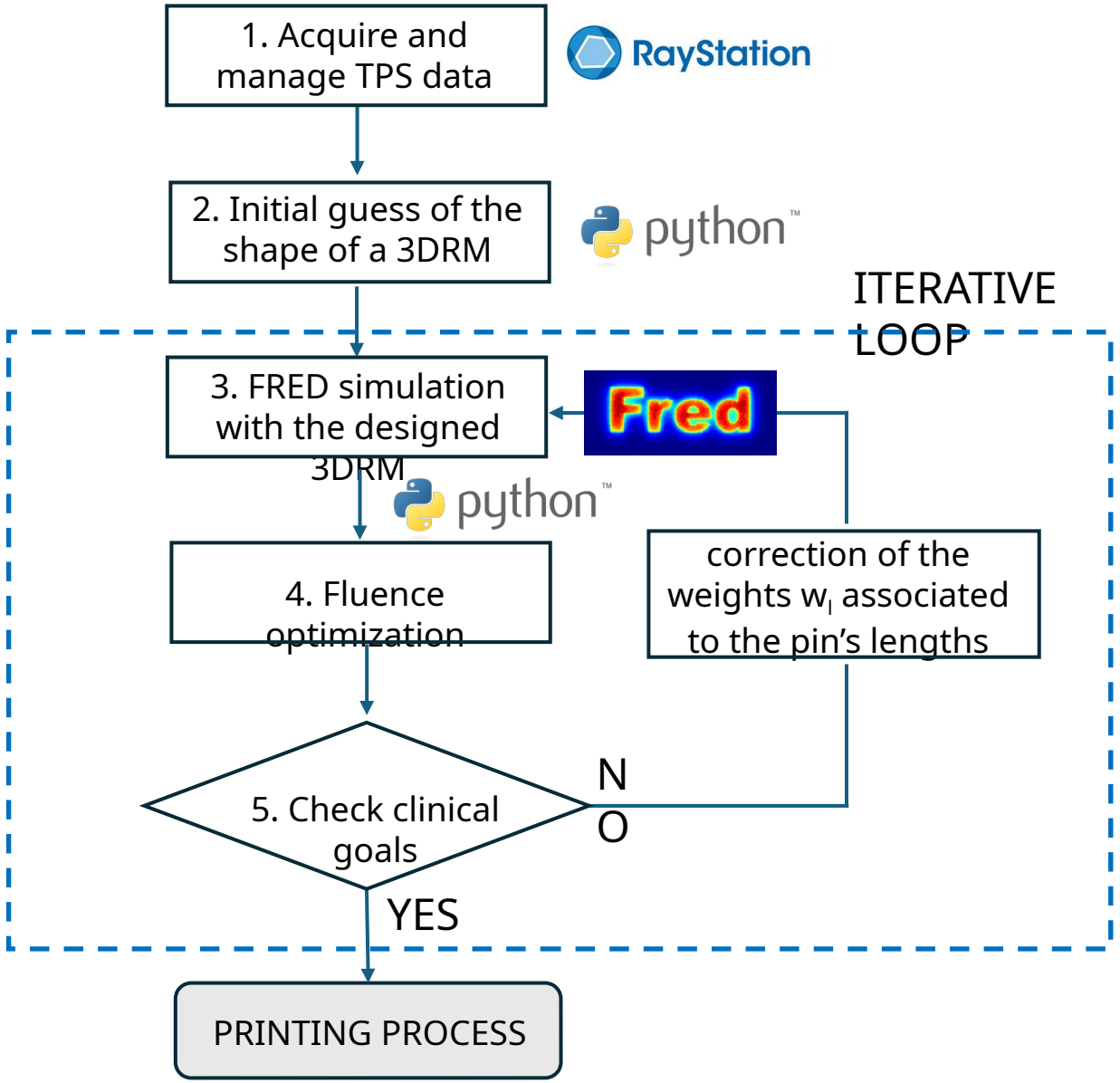
Development of an algorithm to correct the pin's weights based on the dose distribution
(e.g. matching algorithm)





Time estimate:
(10s (FRED) + 1min (fluence opt)) x 100 iter =
≈ 1-2 hours

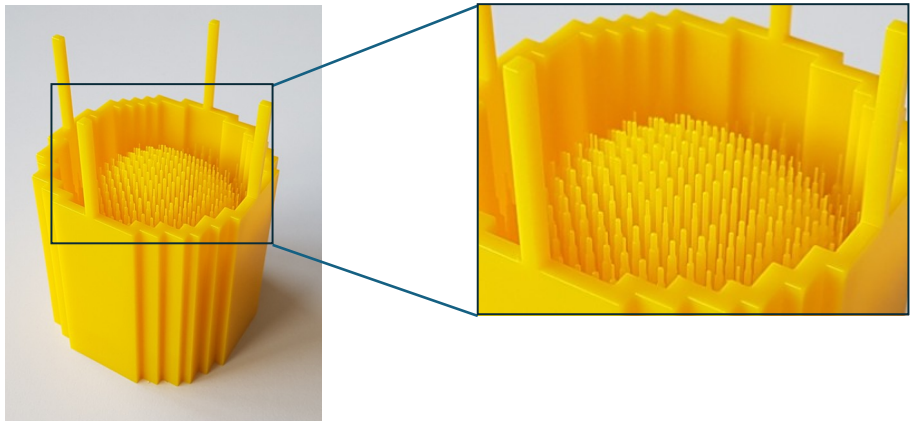
3DRM DESIGN


1. Acquire and manage TPS data
RayStation
2. Initial guess of the shape of a 3DRM
python
ITERATIVE LOOP
3. FRED simulation with the designed 3DRM
Fred
python
4. Fluence optimization
correction of the weights w_l associated to the pin's lengths
5. Check clinical goals
NO
YES
PRINTING PROCESS

PRINTING & TESTING

PRINTING

- 3D printing performed with a stereolithographic printer in Vero resin (Stratasys™) at UniTN-DICAM



R. Cristoforetti (M. Thesis), 2022

TESTING

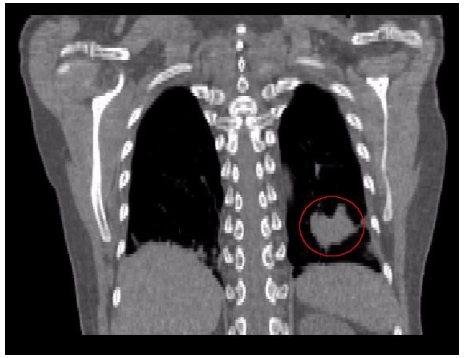
Perfomed at the Proton Therapy Center of Trento (APSS):

- Experimental room
  - To test first 2DRM prototypes
- Gantry
  - Through the fabrication of a dedicated support and alignment system for the 3DRM (ProM™)
  - with dosimetric Tests and QA



OUTLOOK

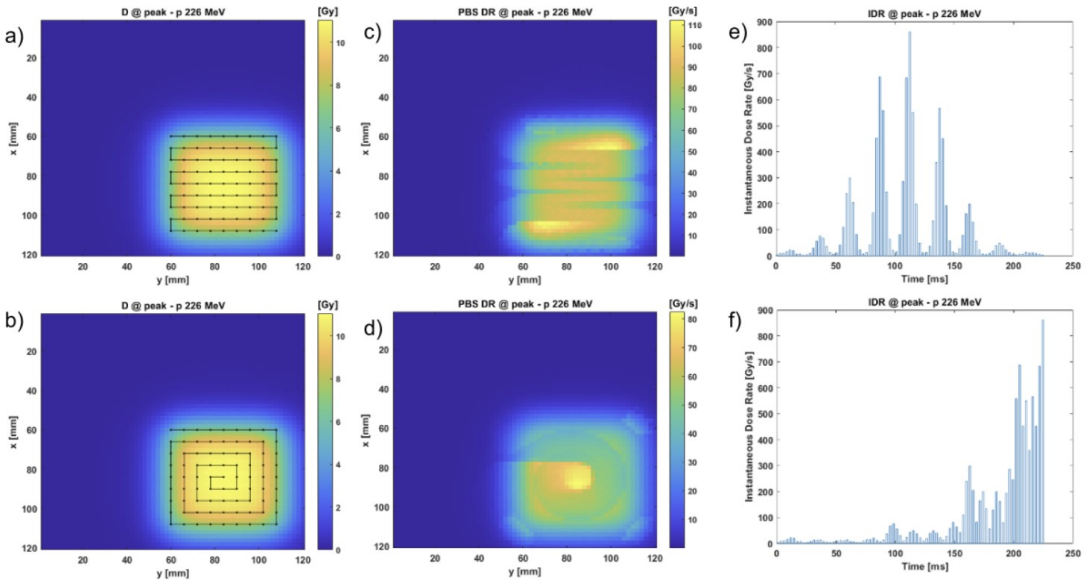
FORM-3D aims to make a concrete step towards the translation of FLASH proton therapy into clincs, in the framework of the PTC of Trento, to be possibly extended to other facilities



(but not only)

A part from Flash Radiotherapy, the project will be important also:

- To speed up the current PT treatments, with an impact in the treatment of moving targets (e.g. Lung) and to better allocate the PTC resources
- Create the basis for the use of FRED as an independent MC tool at the PTC of Trento for further research
- Fast calculation of dose rate metrics with biophysical implications
- Contribute to the development of FRED, due to a ongoing collaboration
- Possible technological transfer



COSTS

Table 3: costs of the project per year

Primo anno		
Missioni	Meeting Roma1/Roma3 per sviluppo algoritmo	2.5k€
Consumabili	Materiali per stampa 3D	10k€ *
Inventariabili	Acquisto workstation GPU	10k€
Servizi	Costo turno fascio	5k€ (SubJudice all'approvazione del PAC)
	Progettazione/realizzazione supporto 3DRM (ProM)	20 k€ *
Tot primo anno		47.5 k€
Secondo anno		
Missioni	Meeting Roma1/Roma3 per sviluppo software	2.5k€
Consumabili	Materiali per stampa 3D	10k€ *
Servizi	Costo turno fascio	2.5k€ (SubJudice all'approvazione del PAC)
Tot secondo anno		15 k€
Totale		**62.5 k€**

GANTT & WORKPACKAGES

Milestones:

WP1.1: Development of an optimization algorithm for 2DRM
WP1.2: Optimization of the 3DRM algorithm for simple geometries
WP1.3: Optimization of the 3DRM algorithm for patient specific cases
WP1.4: Implementation of other metrics in the optimization algorithm (DADR, PBDR)

WP2.1: Testing of 2DRMs in the experimental room
WP2.2: Design and production of a mounting system for the gantry
WP2.3: Testing of 3DRMs for simple geometries in gantry
WP2.4: Testing of patient specific 3DRMs in gantry

First year	M.1	M.2	M.3	M.4	M.5	M.6	M.7	M.8	M.9	M.10	M.11	M.12
Study of the literature on 3DRM algorithms												
Development of an optimization algorithm for 2DRM					1							
Printing of 2DRMs												
Testing of 2DRMs in the exp room							1					
Optimization of the 3DRM algorithm for simple geometries												
Printing of simple geometry 3DRMs												
Testing of 3DRMs for simple geometries in gantry												
Design and production of a mounting system for the gantry												2
Second year	**M.1**	**M.2**	**M.3**	**M.4**	**M.5**	**M.6**	**M.7**	**M.8**	**M.9**	**M.10**	**M.11**	**M.12**
Optimization of the 3DRM algorithm for simple geometries		2										
Optimization of the 3DRM algorithm for patient specific cases						3						
Implementation of other metrics in the optimization algorithm (DADR, PBDR)										4		
Testing of 3DRMs for simple geometries in gantry			3									
Printing of patient specific 3DRMs												
Testing of patient specific 3DRMs in gantry												4

ORGANIZATION

Name	Position	Unit	FTE
Alberto Taffelli*	PostDoc researcher	TIFPA	0.8
Francesco Tommasino	Associate Professor	TIFPA	0.2
Emanuele Scifoni	Primo Ricercatore	TIFPA	0.1
Francesco Cordoni	Associate Professor	TIFPA	0.1
Marco Battestini	PhD student	TIFPA	0.2
Andrea Attili	Primo Ricercatore	ROMA 3	0.2
Angelo Schiavi	Associate Professor	ROMA 1	0.2
Francesco Fracchiolla	Medical Physicist	APSS	0.2
Stefano Lorentini	Responsible of Medical Physics in ProtonTherapy	APSS	0.2
Totale			**2.2**

* principal investigator

FORM-3D:
Fast **O**ptimization of **R**ange **M**odulators **3D**

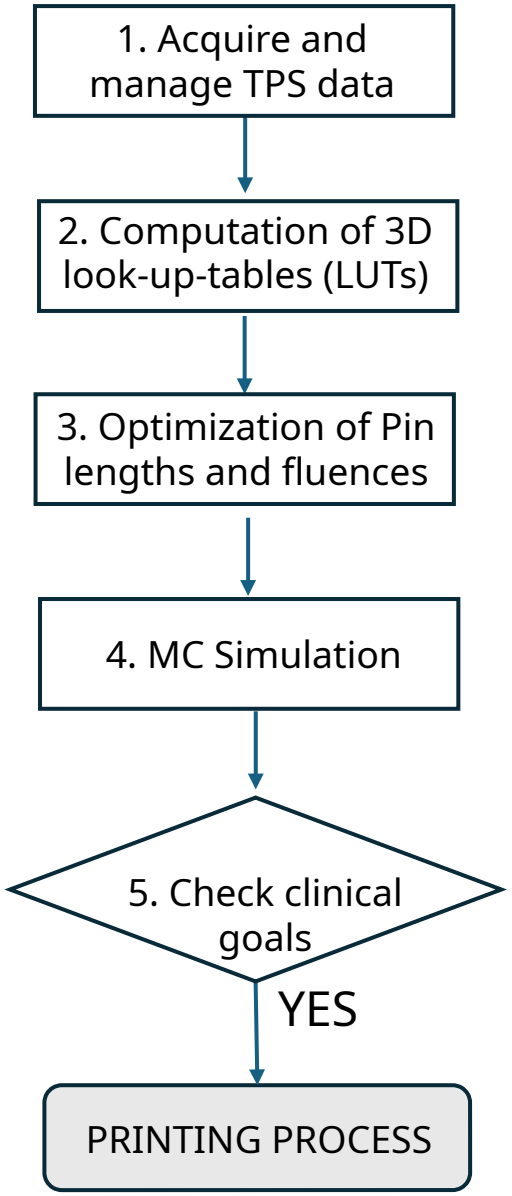
Grant Giovani INFN, November, 19th, 2024

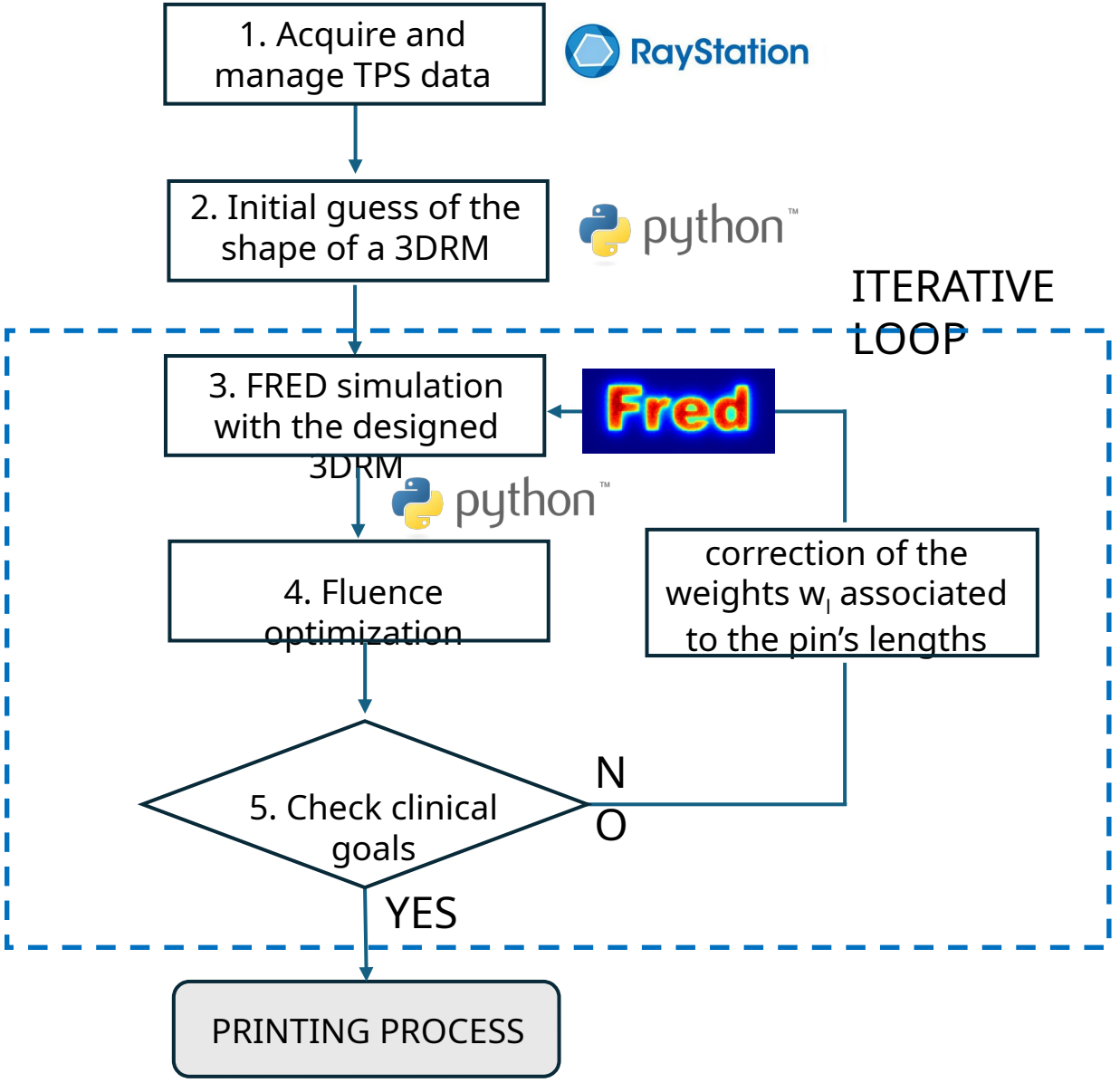
Alberto Taffelli

Rome, November, 19th, 2024

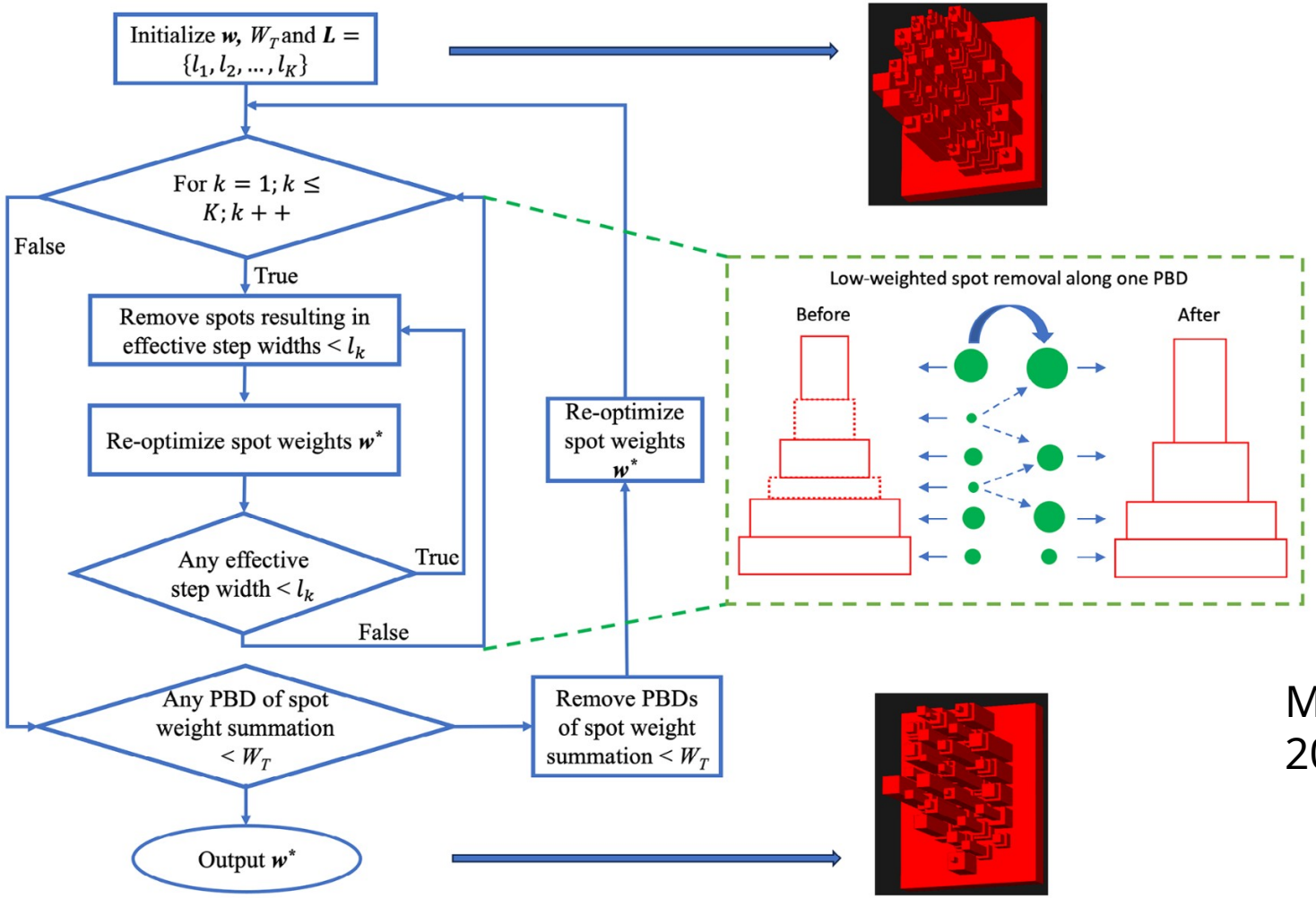
BACKUP SLIDES

3DRM DESIGN


1. Acquire and manage TPS data
2. Computation of 3D look-up-tables (LUTs)
3. Optimization of Pin lengths and fluences
4. MC Simulation
5. Check clinical goals
YES
PRINTING PROCESS


1. Acquire and manage TPS data
RayStation
2. Initial guess of the shape of a 3DRM
python
ITERATIVE LOOP
3. FRED simulation with the designed 3DRM
Fred
python
4. Fluence optimization
correction of the weights w_l associated to the pin's lengths
5. Check clinical goals
NO
YES
PRINTING PROCESS

3DRM DESIGN



Ma et al. 2023

FIGURE 2 Scheme of the nested PBD-based iterative spot reduction process. The green dashed box highlights an example of low-weighted spot removal for one PBD within one iteration of the inner loop. The green dots represent spots, with their sizes proportional to the spot weights. The resulting ridge pins before and after the spot removal are outlined in red. The corresponding steps of the ridge pin to the removed spots are contoured in dashed lines. Examples of the resulting pin-RFs for a field before and after the nested PBD-based iterative spot reduction process are also displayed.

3DRM DESIGN

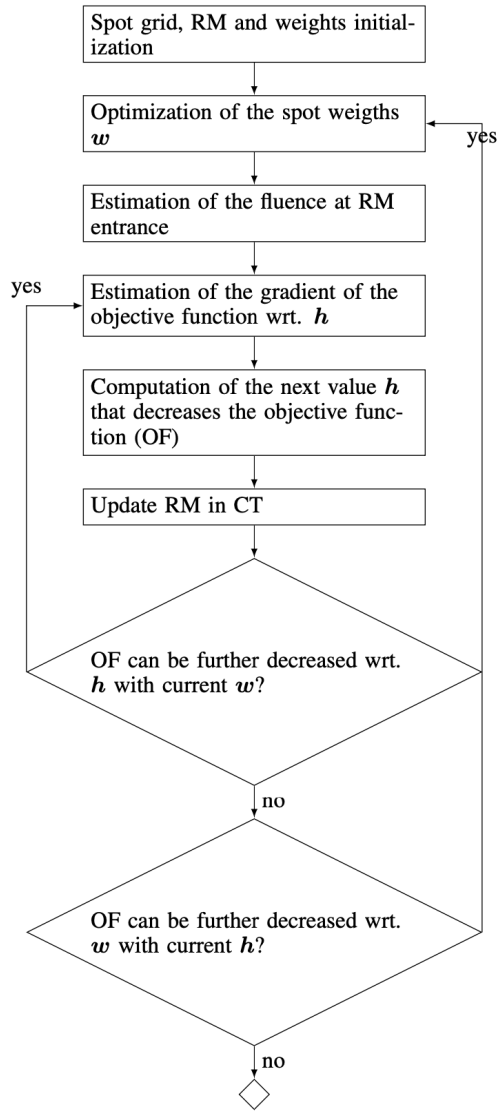


Figure 2: Schematics of the iterative optimization of the individual heights of the range modulator (RM) named h and the PBS spot weights named w.

Deffet et al. 2023

DADR PBDR metrics

$$DADR_i = \sum_{j=1}^{n} \frac{(d_{ij}w_j)(d_{ij}BI_j)}{\sum_{j=1}^{n} d_{ij}w_j}$$

$$DR_i^{PBS} = \frac{D_i - 2d_t}{T_i}$$

